

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculator is allowed.
3. Figure to right indicates full marks.
4. Assume suitable data wherever necessary & mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define the following terms i. Unit cell ii. Atomic packing factor iii. Co-ordination number	Remember	6
B) Sketch the following crystal imperfections i. Vacancy ii. Interstitialcy iii substitution impurity	Remember	6
C) Draw stress strain diagram for mild steel, Show effect of carbon content on shape of stress strain diagram.	Understand	6
Q. 2 Solve Any Two of the following.		12
A) What is Solid Solution? Differentiate between substitutional and interstitial solid solution?	Remember	6
B) Draw neatly iron carbon-carbon equilibrium diagram and mention the three variant reactions involved in the phase diagram.	Understand	6
C) What is T-T-T diagram? How it is different from phase diagram? Explain.	Understand	6
Q. 3 Solve Any Two of the following.		12
A) What is annealing? List different types of annealing along with their purpose.	Remember	6
B) Explain with neat sketches different types of flame hardening.	Remember	6
C) Write short notes on the following with their objectives. i. Normalizing ii. Tempering	Understand	6
Q. 4 Solve Any Two of the following.		12
A) Explain steps in specimen preparation for microscopy.	Remember	6
B) Explain the principle of working of optical metallurgical microscope. Compare it with electron microscope.	Understand	6
C) For spark test draw the sparks for the following specimen. i CI ii MS iii HSS	Remember	6
Q. 5 Solve Any Two of the following.		12
A) Write a note on strain hardening.	Understand	6
B) Explain Dye penetrant test. What are its Applications?	Remember	6
C) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples.	Understand	6